Compute locally, publish automatically

How QCArchive makes sharing a side effect of running the calculation

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Lori Burns maintains Psi4, an open-source quantum chemistry package, and has worked for ten years with the Molecular Sciences Software Institute (MolSSI) as an associate on QCSchema, a shared format for quantum chemistry calculations, and QCArchive, the software and public database built on it. The public servers now hold about 150 million records.[1]

Provenance is required; a name is not

QCSchema has been developed with the community since about 2017. The part that matters for re-use is a required provenance section recording the program that produced a result, its version, and the routine used; people can add runtime conditions, the machine architecture, how long the job took, and anything else they choose. A person's name is not part of it. In a QCSchema file the only trace of a person is incidental, a username or file path echoed in the human-readable output. On QCArchive the record is tied to the account and project that submitted it, and that cannot be separated from the record.

Cite the paper, not the data

Attribution works through papers: the people who generate a dataset write one, and users cite its DOI. The data themselves do not have DOIs. She has met more people who want to know what license is on data they might use than people asking what license to put on their own. Awareness of licensing has grown and people try to use only what they are allowed to, but the data have not kept up. There is no license on QCArchive data. MolSSI's general view is that chemistry data is not copyrightable, though it is still licensable. The Basis Set Exchange, a collection of hundreds of published basis sets, each hundreds of numbers per element, carries no license for the same reason. The software is a different matter: Psi4 began under the GPL and moved to the LGPL nine years ago so that companies could use it, and the distributed package does not contain GPL code.

Immediate availability if you decide to make data public

A user submits a specification (a method, a basis set, a batch of molecules) to a QCArchive server, which hands the jobs to compute the user provides, usually a university cluster; the cluster runs them, attaches the provenance, and returns the compact QCSchema records. If the server is MolSSI's, the records are public the moment they land, with no embargo. If it is the user's own, which the same free software supports, nothing leaves the group. The Sherrill group runs in-progress research privately this way. There has been no pushback, because the private route exists, and we see setting up the infrastructure as the main barrier to sharing: a chemist with a list of molecules can write a twenty-minute script that runs everything locally, and that is exactly what most people need. QCArchive adds storage, deduplication, throughput, error correction, and multi-machine runs, but setting it up takes time and knowledge. Once a group has it, the next project is easy, but many never take the first step.

Coordinating with a power user was a good move

Coordinating early with a downstream user who needed the data at scale was, in her view, the thing MolSSI got most right. Open Force Field, an industry–academic consortium with industry funding and its own scientific staff, runs all its calculations through QCArchive. Fitting a force field means driving selected single bonds in a molecule through its rotation and optimizing the geometry at every step, which takes hundreds to thousands of quantum chemistry gradients per bond; QCArchive decomposes each request into those jobs, runs them, and stores the results in a queryable database. QCArchive has no funding of its own, and MolSSI has sought it several times without success; a power user depending on it at that scale makes a clear case for continuing the work.

[1] Reference implementation of QCSchema: `https://github.com/MolSSI/QCElemental/tree/master/qcelemental/models/v2`.

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